

Brain Computer Interfaces for Enhancing Motor Imagery Classification Using EEG Signal

Rajveer Singh Lalawat

Department of ECE

PDPM-IITDM

Jabalpur, India

rajveer.lalawat@gmail.com

Varun Bajaj

Department of ECE

MANIT Bhopal

Bhopal, India

varunb@iiitdmj.ac.in

Prabin Kumar Padhy

Department of ECE

PDPM-IITDM

Jabalpur, India

prabin16@iiitdmj.ac.in

Abstract—The aim of signal processing with transformation techniques, is to precisely identify frequencies and extract minute signal variations. Previous studies have employed various time-frequency distributions (TFD) to extract features from electroencephalogram (EEG) signals, and nevertheless, there is still space for improvement. This article introduces a sophisticated method for the classification of motor imagery (MI) EEG signals. The objective of this investigation is to generate TFD plots by utilizing pre-proceed data with continuous wavelet transform (CWT). These plots are designed to assist in the identification of the features of each MI EEG class. Subsequently, these plots are applied to the classification of ResNet-20, ResNet-50, and Long Short-Term Memory (LSTM) models. The LSTM outperforms the ResNet models, as evidenced by experiments. Our results indicate that this method enhances the efficacy of the brain-computer interface (BCI) for MI tasks and provides substantial advantages for biomedical applications.

Keywords—Brain-Computer Interface (BCI), Motor Imagery (MI), electroencephalogram (EEG), Time-Frequency Distribution (TFD), Classification.

I. INTRODUCTION

In the current era, neurologists can identify the brain wave with significant assistance from EEG signals. As EEG signals are non-stationary and complicated, it might be challenging to extract the relevant information. The BCI, a technological that creates a system that eliminates the need for any physiological movement by eliminating the need for peripheral nerves and muscles. An individual creates MI-based EEG signals by cognitively imagining an activity without actually performing it [1]. MI-BCI allows humans to exercise their autonomy and aids in recovery. BCIs significantly contribute to the diagnosis and surveillance of many illnesses these are looking at brain activity, BCIs provide an interesting study of neurological diseases such as epilepsy, Parkinson's, and stroke [2]. By recognizing patterns of brainwaves connected with mental health conditions including anxiety and attention deficit hyperactivity disorder (ADHD), BCIs assist in diagnosing and monitoring corresponding treatments. BCIs evaluate abnormal sleeping patterns in sleep disorders like insomnia and sleep apnea. BCIs can assist to track brain injury, evaluate cognitive performance, and identify dementia. BCIs provide potential techniques for better clinical treatment and rehabilitation by assessing motor

and sensory deficits caused by conditions such as brain injuries and amyotrophic lateral sclerosis (ALS). The EEG based MI signals [3], [4] is one of the most practical approaches to consider BCI. These interfaces provide a significant degree of adaptability. Those who have motor problems may engage with the technology via the several types of motor imagery exercises [5] [6].

Gupta et al. suggested the FAWT method based on temporal domain features, kNN classifier, and EEG data to categorize MI activity [7]. Slow cortical potentials (SCRs) were classified in [8] using WPA with MLP. Utilizing EMD and Hilbert transform, Taran et al. proposed to produce IMFs for LS-SVM classifiers [9]. Bhattacharyya et al. claim that Fourier Bessel Series Expansion (FBSE) might improve empirical wavelet transform (EWT). Wavelet-based filter banks separated signals into narrow-band components; the normalized Hilbert transform was then used to evaluate instantaneous frequency and amplitude envelope with reduced MSE [10]. Zhou et al presented a new technique using LSTM classifier and wavelet envelope analysis [11]. Authors examined EEG-based BCI applications using deep learning methodologies and evaluated contemporary models for motor imagery attention recognition, emphasizing their benefits and drawbacks [12]. Authorial evaluations of EEG-based brain-computer interfaces, emphasizing MI applications using wearable sensors. Evaluate technical and computational achievements by reviewing 84 publications, while establishing benchmarks and theoretical frameworks for future improvements [13].

This study introduces an BCI-MI with CWT technique tailored for multi-MI across four distinct phases. The open-source multi-class EEG dataset was partitioned into four-second segment epochs [14], each comprising 1000 samples, with a sampling frequency of 250 Hz. The following important contributions are made by this paper:

- The developed framework for automatic identification four stage of motor imaging classes.
- This novel method with CWT provides a detailed analysis of EEG signals in both time and frequency domains, making it useful for capturing the dynamic changes in EEG data.
- The CWT extracting key features from the CWT scalogram of BCI-MI analysis efficiently and three models

utilized to improve our system robustness.

The article's remaining sections are organized as follows: Section II briefly describes the multi-class MI system and introduces the paper's database. In Section III, we step through the results and discussion of the proposed MI-based system. Section IV concludes and summarizes the work including future work.

II. METHODOLOGY

The developed framework comprises three main steps. Firstly, individual subject data undergoes thorough comprehensive analysis. Secondly, the time frequency distribution plot generated that is call scalogram is obtained via CWT, followed by identification of all local features of each classes. This

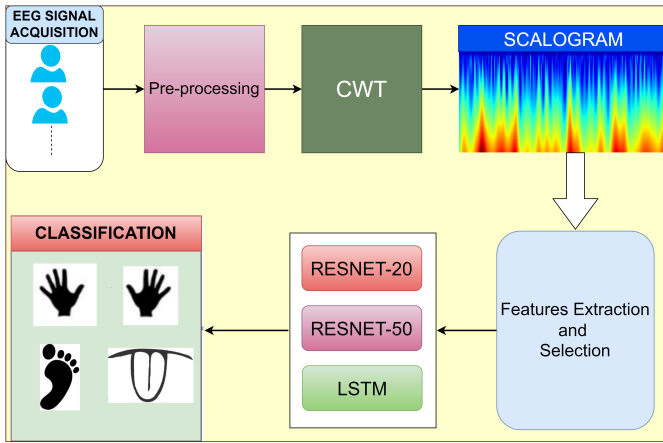


Fig. 1. Schematic representation of the proposed Framework

step is crucial in enabling flexibility in signal analysis and accurately estimating individual frequencies present. Finally, classification is performed using LSTM with high accuracy and also compared with other well know models. The proposed approach shows in Fig. 1.

A. Dataset

The present work utilizes data obtained from the BCI Competition IV 2a, specifically EEG recordings of nine individuals participating in several cognitive activities [15]. The dataset has 1000 samples, each lasting 4 seconds, and running at a sampling rate of 250Hz, distributed across multiple epochs. In all, twenty-two electrodes composed of Ag/AgCl (Channels) were used to capture the EEG data, with an inter-electrode spacing of 3.5 cm. Each individual participants completes seventy-two epochs consisting of four cognitive states.

Every participant undergoes 72 epochs, which include four cognitive states. This work uses 5th-order Butterworth band-pass filter with a lower (upper) cutoff frequency of 0.5 (38)Hz was then applied to improve the signal quality and reduce noise and unwanted signals. An illustrative example based on multi-stage MI signal shown in Fig. 2.

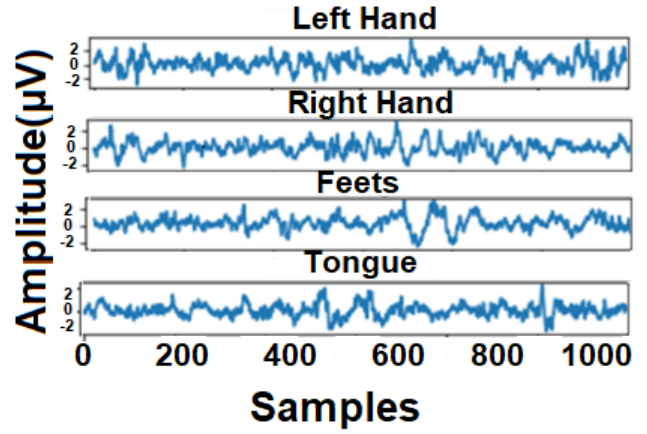


Fig. 2. Examples of EEG Signals for Different MI Stages

B. Continuous Wavelet Transform

The Continuous Wavelet Transform (CWT) provides a continuous view in the time-frequency domain, allowing non-stationary data to be analyzed. The CWT linearly translates a signal along the time axis after converting it into a sequence of continuous wavelet functions at different sizes. At various scales and time periods, the system determines if each wavelet function adequately represents the signal. As a result, a time-frequency representation allows one to examine how the signal's frequency component evolves with respect to both time and scale [16]. Wavelets are a kind of bandpass filter that have finite energy, a zero-average function, and regular oscillations. When it comes to frequency and time sector localization, the CWT outperforms the Fourier transform. If we use the CWT on the signal $x(t)$, the mathematical statement becomes:

$$CWT_{a,b}[w(t)] = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t) \cdot \psi^* \left(\frac{t-b}{a} \right) dt \quad (1)$$

The CWT coefficient at scale a and translation b is represented by the item $CWT_{a,b}$. (The first signal variable is represented by $x(t)$). We derive the mother wavelet function from $\psi(t)$. Representing as a , the scaling factor determines how the wavelet quantity is stretched or compressed. The sign b denotes temporal displacement or translocation. Represented as $\psi^*(t)$ the complex conjugate of the mother wavelet function. The frequency components of the signal are found using the scale factor a ; smaller values of a indicate higher frequency components and larger values a indicate lower frequency components. By guiding the wavelet along the time axis, the translation parameter b lets frequency components be localized in time. The CWT allows one to do an extensive investigation of the spatial and temporal variations in the frequency content of a signal. Whereas the parameters a determine its dilation and compression, the transformation parameter b allows the temporal displacement of the wavelet. This approach helps the CWT to become a great instrument for analyzing time-frequency signals.

C. Feature Extraction

The CWT extracts extensive EEG characteristics to categorize motor imagery. CWT creates a scalogram that displays why EEG signal frequency changes over time by breaking it down into its time-frequency components. Anyone may learn important things from this scalogram. These traits illuminate the brain's complex pattern of activity while motor imagining. The method appropriately categorizes MI states utilizing LSTM time modeling and the CWT scalogram's high feature set. In brain-computer interface (BCI) systems, CWT-based extraction of features and LSTM class improve motor imaging job discrimination accuracy and robustness.

D. Deep learning Models

This paper provides a detailed description of the classification algorithms—ResNet20, ResNet50, and LSTM—along with a comparative examination of their respective benefits and shortcomings. ResNet20, a 20-layer residual neural network, employs skip connections to mitigate the vanishing gradient issue in deep architectures. It comprises residual blocks with convolutional layers, batch normalization, and ReLU activations. ResNet20 is effective for small datasets but has limited ability to manage complicated data because to its shallow architecture. In contrast, ResNet50, a richer form with 50 layers, incorporates bottleneck blocks (1x1, 3x3, 1x1 convolutions) enabling it to manage bigger datasets and more complex jobs by capturing deeper feature hierarchies. Nevertheless, the more complex design entails greater processing demands and an elevated danger of overfitting.

Unlike ResNet structures, LSTM is specifically designed for sequence data and is adept at capturing temporal relationships, making it especially effective for EEG signal categorization. The LSTM model has memory cells equipped with forget, input, and output gates, enabling it to save and regulate information throughout extended sequences. LSTM is extremely successful for time-series prediction but is more computationally intensive than ResNet models and may have vanishing gradient issues in lengthy sequences, however this is alleviated by its internal gating mechanisms.

1) *ResNet-20*: Architecture: ResNet-20 has 20 layers, making it a comparatively shallow variant of the ResNet architecture. It is mostly intended for picture classification jobs. Residual connections are It employs skip connections (or residual connections) that facilitate the transfer of gradients through the network without disappearing, hence allowing the successful training of deeper networks. ResNet-20 demonstrates comparable performance on benchmark datasets such as CIFAR-10 while maintaining computational efficiency owing to its reduced size. It is often used in educational contexts and serves as a foundational model in research because of its simplicity and efficacy in illustrating the advantages of residual learning.

2) *ResNet-50*: ResNet-50 is an enhanced variant of the ResNet architecture, with 50 layers, and is extensively used for intricate images classification tasks, including those in ImageNet. This architecture utilizes a bottleneck design in

its building blocks, minimizing the amount of parameters while preserving good performance with the implementation of three layers per bottleneck. ResNet-50 attains state-of-the-art performance on many bio-medical images classification benchmarks, markedly surpassing conventional designs owing to its depth and residual connections. It is a widely favored option in practical applications and competitions, functioning as a foundational model for several computer vision tasks, such as object identification and segmentation.

3) *Long Short-Term Memory (LSTM)*: The network is use of memory cells with learning mechanisms, that are well designed to allow for the updating and selective the data information over time. due to this, LSTM is useful for that condition because of knowledge in temporal patterns, such as EEG signals classification. To learning the temporal dynamics behaviorism in EEG signal with the LSTM when it comes to EEG signal, particularly for MI classification. To utilizes with memory cells to store previously processed data, the LSTM analyzes EEG signals sequential properties. The model is learn to initial most significant parts of the signal and ignore the rest data sequence, so thanks to the gating mechanisms, that include input, output, and forget gates.

A strong system for decoding MI classification, LSTM improves the performance of BCI applications by capturing both short-term fluctuations and long-term dependencies in EEG signals. The cell state c_t is a memory component that carries information through time steps. It is updated using the previous cell state c_{t-1} and the new information from the input at time t . The update is governed by the forget gate f_t , the input gate i_t , and the cell state candidate $\tilde{c}_t$. The update equation is:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (2)$$

The forget gate f_t determines which parts of the previous cell state should be retained. It is computed using the sigmoid activation function applied to the previous hidden state h_{t-1} and the current input x_t , combined with a bias term b_f . The equation is:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (3)$$

The input gate i_t controls which new information is added to the cell state. It uses the sigmoid function to process the current input and the previous hidden state. The equation is:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (4)$$

The cell state candidate $\tilde{c}_t$ is generated from the current input x_t and the previous hidden state h_{t-1} , using a hyperbolic tangent (tanh) function. This candidate represents the potential new information to be added to the cell state. The equation is:

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (5)$$

The output gate o_t determines the portion of the updated cell state that will be exposed as the hidden state. It is computed using the sigmoid function, and the hidden state h_t is then

obtained by applying the output gate to the cell state, combined with a tanh function. The equations are:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (6)$$

$$h_t = o_t \odot \tanh(c_t) \quad (7)$$

To summarize, LSTM systems make use of the following parameters the forget gate (f_t), the input gate (i_t), the cell state candidate ($\tilde{c}_t$), and the output gate (o_t) to efficiently handle and remember data across extended sequences. LSTM networks are great at time series prediction and sequence classification because of the network's ability to learn complicated temporal patterns and relationships in sequential input using a mix of gates and cell state updates.

III. RESULTS AND DISCUSSION

The purpose of this research is to develop an effective system of multi-class motor imagery stage identification using BCI-IV data-set. The Fig. 3. shows the CWT generated scalogram

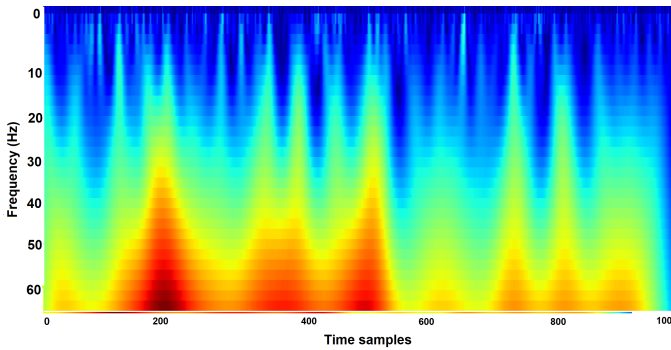


Fig. 3. Example of the TFD-CWT segment plots with Left class

gram EEG signal segment from the original left hand stage EEG signals. similarly, we generated all remain classification stages with the same approach. Generate the scalogram plot for that specific sample in that particular scenario, with a length of 1000 samples. Therefore, by using 1000 samples, each representing a 4-second epoch, we can ensure a comprehensive analysis of MI data with sufficient temporal resolution to capture relevant patterns and features within each epoch.

Figure 4. shows the unaltered EEG signal training accuracy from the BCI-MI EEG signals. Training accuracy is increase from low to high value indicate that out system in doing well with respect to our data. The model is train for 100 epochs after that over fitting has been occurred. In this study, we implemented a deep learning framework using PyTorch to classify motor imagery (MI) EEG signals, which were pre-processed and CWT to generate time-frequency domain representations. These CWT-based images were then treated as input to the model. The dataset was organized into sub directories for each class, and split into training (80%) and testing (20%) sets using PyTorch random split. We applied basic data augmentation, resizing the images to 224x224 pixels to fit the ResNet architecture. The output layer was adjusted to predict the four EEG MI classes.

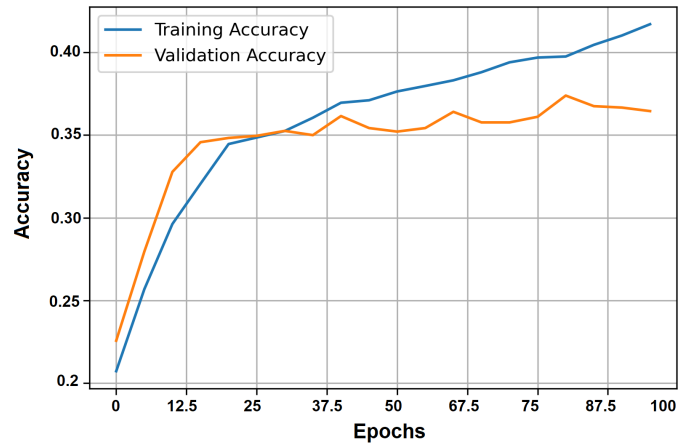


Fig. 4. Training accuracy and validation accuracy with epochs using LSTM Model

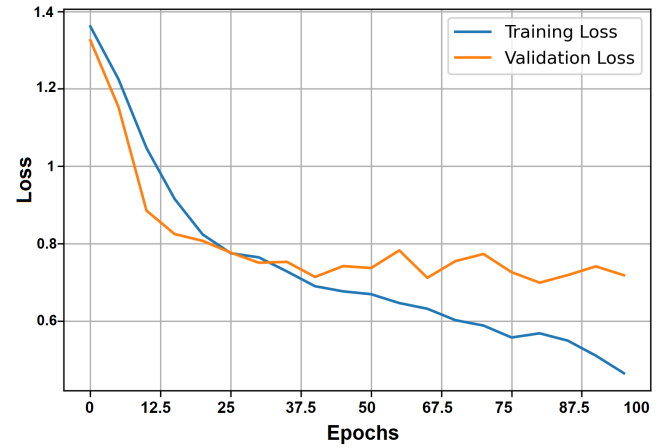


Fig. 5. Training Loss and validation loss with epochs using LSTM Model

Figure 5. shows the unaltered EEG signal training loss from the BCI-MI EEG signals. Training loss is decreases from high to low value indicate that out system in doing well with respect to our data. The model is train for 100 epochs after that over fitting has been occurred. we used test accuracy along with precision, recall, and F1-scores to assess the classification performance across all classes. Additionally, we generated a confusion matrix to visualize the model's performance in distinguishing between the different classes. The results showed a gradual improvement in accuracy throughout training. Notably, although the ResNet-20 model achieved competitive results, earlier tests with LSTM demonstrated superior classification performance, particularly for motor imagery EEG tasks. Ultimately, the trained model for future use, and the confusion matrix provided valuable insights into classification challenges between EEG classes. In Fig. 6. shows the Receiver Operating Characteristic (ROC) curve showing the Area Under the Curve (AUC) for Class 1, highlighting the model's performance in distinguishing Class 1 from the other classes. Our model attained an AUC (Area Under the Curve) score of 0.62. This number indicates that the model has a modest capacity

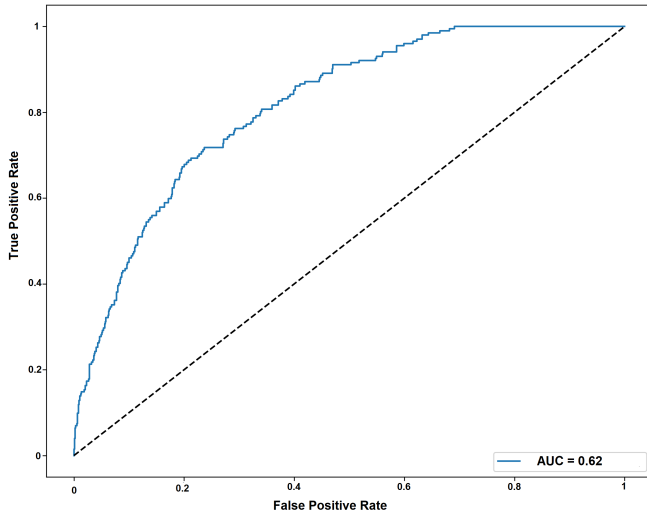


Fig. 6. ROC curve illustrating the AUC for Class 1, demonstrating the model's effectiveness

to differentiate between the classes for the specified motor imagery (MI) EEG data. An AUC of 0.62 indicates a 62% probability that the model would accurately rank a randomly selected positive case above a randomly selected negative instance. The AUC value indicates that the model outperforms random guessing (AUC of 0.5), although it implies that further improvements in model performance are necessary. Optimizing the model, extracting features, or experimenting with other topologies may enhance classification accuracy and AUC in subsequent tests. The LSTM model for EEG signal categorization, with two layers, each containing 128 hidden units. The model transforms input data, derived from CWT plots, into a sequence of 224 items at each step. A fully linked layer succeeds the LSTM layers to forecast the output classes. The model was trained with the Adam optimizer with a learning rate of 0.001 and Cross Entropy Loss as the loss function. The batch size was set to 32, and training was performed across 100 epochs. Metrics including accuracy, precision, recall, and F1-score were monitored, and a confusion matrix was examined to assess categorization performance.

TABLE I
PERFORMANCE METRICS FOR LSTM MODEL WITH 10-FOLD
VALIDATION

Folds	Precision	Recall	F1 Score	Accuracy
1	0.416	0.405	0.418	0.410
2	0.422	0.426	0.424	0.436
3	0.429	0.413	0.415	0.410
4	0.438	0.431	0.435	0.426
5	0.428	0.421	0.429	0.425
6	0.418	0.411	0.419	0.416
7	0.429	0.423	0.425	0.430
8	0.438	0.431	0.439	0.436
9	0.418	0.411	0.419	0.425
10	0.428	0.421	0.429	0.436

TABLE II
10-FOLD AVERAGE CLASSIFIER ACCURACY COMPARISON

Classifier	Accuracy (%)
ResNet-20	40.7
ResNet-50	41.2
LSTM	42.57

The performance metrics of the LSTM model during 10-fold cross-validation are shown in Table I, illustrating the model's consistency across different parameter. The results for precision, recall, F1-score, and accuracy show reasonable stability over the folds, with accuracy fluctuating between 0.410 and 0.436. The model has moderate performance, with accuracy and recall values exhibiting modest fluctuations while retaining comparable trends, signifying the model's balanced proficiency in accurately predicting classes while limiting false positives and false negatives. Table II presents a comparison of the mean accuracy of several classifiers with 10-fold validation. The LSTM model had the maximum accuracy of 42.57%, surpassing ResNet-20 at 40.7% and ResNet-50 at 41.2%. This illustrates the superiority of LSTM in processing sequential EEG data, with marginally enhanced classification accuracy compared to ResNet topologies. The categorisation, informed by CWT-derived characteristics, is limited by the capacity to discern unique time-frequency patterns associated with various motor imaging tasks (e.g., hand vs foot movement). Effective classification depends on the selection of pertinent wavelet scales and frequency bands that effectively represent the brain activity disparities between various activities, especially within the alpha(8-12Hz) and beta(12-30Hz) frequency ranges.

This work applies TFD employing CWT to MI EEG signal categorization, adding to BCI signal processing research. EEG signals are decomposed into frequency components by the CWT, giving temporal and frequency domain information. Prior studies had drawbacks, but our strategy improves feature extraction and categorization. This technique has been compared to others for MI-related EEG data processing. Ang et al. [17] classified MI EEG data using CSP and SVM classifiers with good results. They concentrated on spatial filtering without time-frequency analysis. CWT in our work focuses on time and frequency, allowing us to better capture EEG signals' dynamic character. Zhang et al. [18] has been classified MI EEG data using WPD and DNNs. WPD is comparable to CWT, however our technique employing TFD plots and CWT yielded more reliable results, particularly when used with deep learning models like ResNet-20, ResNet-50, and LSTM. Due to its capacity to describe EEG temporal relationships, the LSTM network outperformed ResNet models. This shows that temporal information in time-series EEG data is crucial for classification accuracy. Lawhern et al. [19] has been used EEGNet, a lightweight CNN for EEG classification. The EEGNet work with less parameters, however our research shows that deeper architectures like LSTM, paired with rich CWT characteristics, convolutional models on MI tasks.

ResNet-20 provides simple and quicker training; never-

theless, it may encounter difficulties with intricate datasets, resulting in diminished accuracy. ResNet-50, because to its deeper design, identifies complex patterns and attains superior accuracy; yet, it requires more computer resources and extended training durations. LSTM is proficient at learning temporal dependencies in sequential data, rendering it suitable for tasks such as EEG analysis; yet, its complexity and prolonged training time may pose disadvantages. The selection of these models is contingent upon the particular job demands and the resources at hand.

IV. CONCLUSION

This work presents a method that is for the MI classification, using EEG signal processing methods to improve the precision and effectiveness of BCIs. We used TFD and CWT to find and visualize the frequency components in EEG signals, that is important for differentiating between four MI classes. The CWT for feature extraction with TFD plots that accurately depict the differences in EEG signals corresponding to four MI classes. These TFD plots as an effective technique for finding the basic patterns in the signals, therefore enhanced more precise classifications. The use of CWT produces comprehensive time-frequency graphs, facilitating the collection of complex fluctuations in MI EEG data. The use of sophisticated models such as ResNet-20, ResNet-50, and LSTM reveals a substantial improvement in the classification of MI tasks, with LSTM surpassing the performance of the other models. A few drawbacks include the computational intensity of creating CWT graphs and training deep learning models such as ResNet and LSTM, which need substantial resources. The substantial capacity of models such as ResNet and LSTM poses a danger of overfitting, particularly with short datasets, potentially impairing real-world performance.

The Classification parameters of three different models are ResNet-20, ResNet-50, and LSTM networks. The results indicated that the LSTM model superiors with respected to both ResNets models. This advantage is due to LSTM's capacity to capture temporal relationships in EEG signals, due to appropriate for label data that is MI tasks. The results shows that the our system can use the collected features to classification accuracy, that improving the overall efficacy of BCIs. Furthermore, our findings underscore the promise of using sophisticated signal processing techniques in biomedical applications. This work enhances the classification of MI EEG signals, increase the efficacy of BCIs, that assist persons with motor disabilities by intuitive control of external system. This paper highlights the significance of advanced signal processing methods in the progression of BCI with EEG signals.

- 1) The proposed BCI-based system with CWT is enhancing feature extraction from MI EEG signals, leading to improved classification performance.
- 2) The proposed system validated with ten fold cross-validation and LSTM network outperformed ResNet-20 and ResNet-50 for classification of MT task.

- 3) The potential of the proposed techniques to enhance BCIs for MI tasks, offering significant benefits for biomedical applications and real-time systems.

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